International Baccalaureate®
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MARKSCHEME

SPECIMEN

CHEMISTRY

Higher Level

Paper 2

17 pages

Subject Details: Chemistry HL Paper 2 Markscheme

Mark Allocation

Candidates are required to answer **ALL** questions. Maximum total = **[95 marks]**.

1. Each row in the "Question" column relates to the smallest subpart of the question.

2. The maximum mark for each question subpart is indicated in the "Total" column.

3. Each marking point in the "Answers" column is shown by means of a tick (✓) at the end of the marking point.

4. A question subpart may have more marking points than the total allows. This will be indicated by "**max**" written after the mark in the "Total" column. The related rubric, if necessary, will be outlined in the "Notes" column.

5. An alternative wording is indicated in the "Answers" column by a slash (/). Either wording can be accepted.

6. An alternative answer is indicated in the "Answers" column by "***OR***" on the line between the alternatives. Either answer can be accepted.

7. Words in angled brackets ‹ › in the "Answers" column are not necessary to gain the mark.

8. Words that are <u>underlined</u> are essential for the mark.

9. The order of marking points does not have to be as in the "Answers" column, unless stated otherwise in the "Notes" column.

10. If the candidate's answer has the same "meaning" or can be clearly interpreted as being of equivalent significance, detail and validity as that in the "Answers" column then award the mark. Where this point is considered to be particularly relevant in a question it is emphasized by ***OWTTE*** (or words to that effect) in the "Notes" column.

11. Remember that many candidates are writing in a second language. Effective communication is more important than grammatical accuracy.

12. Occasionally, a part of a question may require an answer that is required for subsequent marking points. If an error is made in the first marking point then it should be penalized. However, if the incorrect answer is used correctly in subsequent marking points then **follow through** marks should be awarded. When marking, indicate this by adding **ECF** (error carried forward) on the script. "ECF acceptable" will be displayed in the "Notes" column.

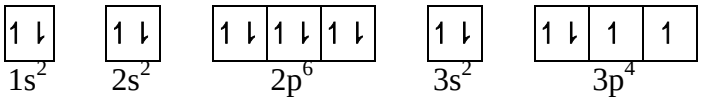
13. Do **not** penalize candidates for errors in units or significant figures, **unless** it is specifically referred to in the "Notes" column.

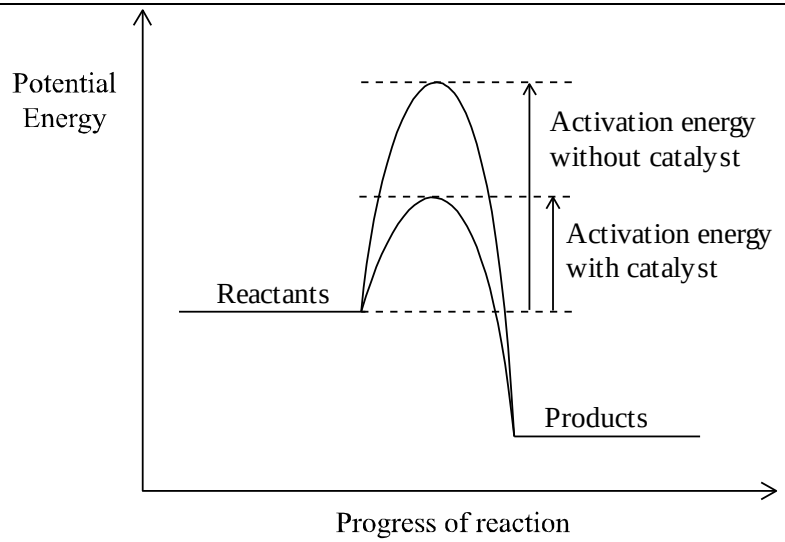
14. If a question specifically asks for the name of a substance, do not award a mark for a correct formula unless directed otherwise in the "Notes" column, similarly, if the formula is specifically asked for, unless directed otherwise in the "Notes" column do not award a mark for a correct name.

15. If a question asks for an equation for a reaction, a balanced symbol equation is usually expected, do not award a mark for a word equation or an unbalanced equation unless directed otherwise in the "Notes" column.

16. Ignore missing or incorrect state symbols in an equation unless directed otherwise in the "Notes" column.

Question			Answers	Notes	Total
1.	**a**	**i**	‹$(22.05+22.15)(0.5)$ =› 22.10 ‹cm^3› ✓		**1**
	a	**ii**	$\left\langle \frac{22.10\times0.100}{1000} = \right\rangle 2.21\times10^{-3}$ / 0.00221 ‹mol› ✓		**1**
	a	**iii**	$\left\langle \frac{0.5\times2.21\times10^{-3}\times1000}{25.00} = \right\rangle 4.42\times10^{-2}$ / 0.0442 ‹$mol\,dm^{-3}$› ✓		**1**
	a	**iv**	‹$4.42\times10^{-2}\times10$ =› 4.42×10^{-1} / 0.442 ‹$mol\,dm^{-3}$› ✓		**1**
	b	**i**	*NaClO*: +1 ‹for chlorine› **and** I_2: 0 ‹for iodine› ✓		**1**
	b	**ii**	ClO^- since chlorine reduced/gains electrons ***OR*** ClO^- since oxidation state of chlorine changes from +1 to –1/decreases ***OR*** ClO^- since it loses oxygen / causes iodide to be oxidized ✓		**1**
	b	**iii**	produces chlorine ‹gas›/Cl_2 ‹on reaction with ClO^-› which is toxic ✓	*OWTTE*	**1**
	b	**iv**	oxidation states are not real ***OR*** oxidation states are just used for electron book-keeping purposes ✓ average oxidation state of sulfur calculated to be +2 ✓ but the two sulfurs are bonded differently/in different environments in thiosulfate so have different oxidation states ✓	*OWTTE*	**2 max**

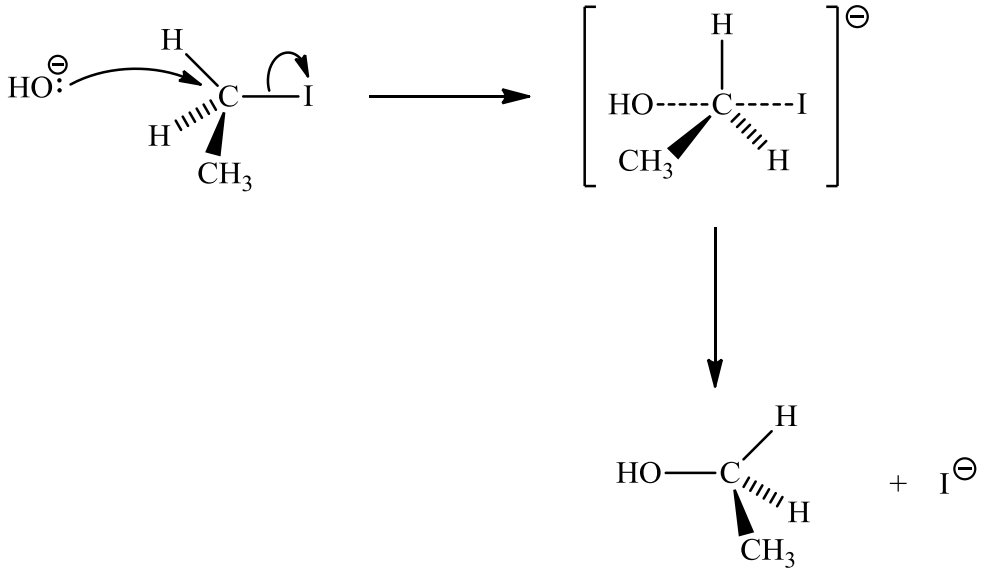
Question			Answers	Notes	Total
	c		*Valid:* addition of oxygen signifies an oxidation reaction so C is oxidized ***OR*** loss of hydrogen signifies an oxidation reaction so C is oxidized ***OR*** oxidation state of C changes from –4 to +4/increases ✓ *Not valid:* loss of electrons might suggest formation of ionic product but not valid since CO_2 is covalent ***OR*** loss of electrons might suggest formation of ionic product but not valid since reaction only involves neutral molecules ✓	*OWTTE*	**2**
	d	**i**	$[Ne]3s^2 3p^4$ ✓	*Electrons must be given as superscript.*	**1**
	d	**ii**	 ✓		**1**

<table>
<tr><th colspan="3">Question</th><th>Answer</th><th>Notes</th><th>Total</th></tr>
<tr><td>2.</td><td>a</td><td></td><td>radical / unpaired electron ✓</td><td></td><td>1</td></tr>
<tr><td></td><td>b</td><td></td><td>
<table>
<tr><th>Change</th><th>Shift</th><th>Reason</th></tr>
<tr><td>Increase in temperature</td><td>LHS</td><td>since ‹forward› exothermic reaction/ $\Delta H < 0$ ✓</td></tr>
<tr><td>Increase in pressure</td><td>RHS</td><td>since fewer ‹gaseous› molecules on RHS ✓</td></tr>
<tr><td>Addition of a catalyst to the mixture</td><td>No change</td><td>since affects rate of forward and reverse reactions equally ✓</td></tr>
</table>
</td><td></td><td>3</td></tr>
<tr><td></td><td>c</td><td></td><td>


correct positions of reactants and products ✓
correct profile with labels showing activation energy with and without a catalyst ✓
</td><td></td><td>2</td></tr>
</table>

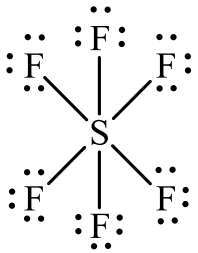
Question			Answers	Notes	Total
	d		$2NO_2(g) + H_2O(l) \rightarrow HNO_3(aq) + HNO_2(aq)$ ✓	*Ignore state symbols.*	**1**
	e		ionic radius of nitrogen is $146\,pm/146\times10^{-12}$ m which is greater than atomic radius which is $71\,pm/71\times10^{-12}$ m due to increased repulsion between electrons ✓	*Values must be given to score mark.*	**1**

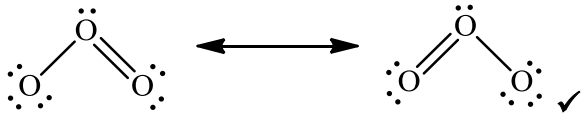
3.	**a**		$HCOO^-(aq) + H_2O(l) \rightleftharpoons OH^-(aq) + HCOOH(aq)$ ✓	*Equilibrium sign must be given for mark.*	**1**
	b		$K_a = 1.8\times10^{-4}$ ✓		**1**
	c		$K_b = \left\langle \frac{K_w}{K_a} = \frac{1.0\times10^{-14}}{1.8\times10^{-4}} = \right\rangle 5.6\times10^{-11}$ ✓		**1**
	d		$K_b = \frac{x^2}{0.12} = 5.6\times10^{-11}$ ✓ $[OH^-(aq)] = 2.6\times10^{-6}$ ‹$mol\,dm^{-3}$› ✓ *Assumption:* $0.12 - x \sim 0.12$ ✓	*Award **[2]** for correct final answer of $[OH^-(aq)]$.* *Accept any other reasonable assumption.*	**3**
	e		$pOH = ‹-\log(2.6\times10^{-6}) =› 5.59$ ✓ $pH = ‹14.00 - 5.59 =› 8.41$ ✓	*Award **[2]** for correct final answer.*	**2**

Question			Answers	Notes	Total
4.	**a**		 curly arrow going from lone pair/negative charge on O in HO^- to C ✓ curly arrow showing I leaving ✓ representation of transition state showing negative charge, square brackets and partial bonds at 180° to each other ✓ formation of organic product CH_3CH_2OH **and** I^- ✓	*Do not allow curly arrow originating on H in HO^-* *Accept curly arrow going from bond between C and I to I in 1-iodoethane or in the transition state.* *Do not allow arrow originating from C to C–I bond.* *Do not award M3 if OH---C bond is represented.* *Inversion of configuration must be shown to score M4.*	**4**
	b		*Rate expression:* $\text{rate} = k[OH^-][CH_3CH_2I]$ ✓ *Molecularity of RDS:* bimolecular ✓		**2**

Question			Answers	Notes	Total
	c		*S_N2:* polar, protic solvents decrease nucleophilic reactivity due to hydrogen bonding ***OR*** polar, protic solvents have a cage of solvent molecules surrounding anionic nucleophile resulting in increased stabilization ‹so are slower› ***OR*** polar, aprotic solvents have no hydrogen bonding so S_N2 reactions are favoured since nucleophiles do not solvate effectively so have an enhanced/pronounced effect on nucleophilicity of anionic nucleophiles ‹so are faster› ✓ *S_N1:* polar, protic solvents favour S_N1 reactions since the carbocation ‹intermediate› is solvated by ion-dipole interactions by the polar solvent ✓		**2**
	d		DMF since aprotic solvent so favours S_N2 ✓		**1**
	e		*A* is indicative of frequency of collisions **and** probability that collisions have proper orientations ✓		**1**
	f		$k = \left\langle \exp\left[\frac{(-87.0\times1000)}{(8.31\times298)} + \ln(2.10\times10^{11})\right] = \right\rangle 1.2\times10^{-4}$ ✓ S_N2 implies second-order so $mol^{-1}\ dm^3\ s^{-1}$ ✓		**2**

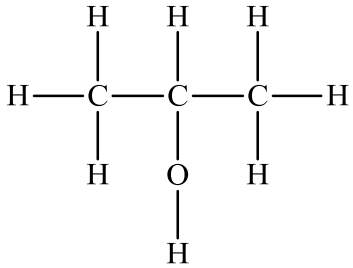
Question			Answers	Notes	Total
5.	**a**		only water/H_2O produced ‹so non-polluting› ✓		**1**
	b		*Bond breaking*: $(1)(H–H) + (4)(C–H) + (1)(C=C)$ ***OR*** $(1)(436) + (4)(414) + (1)(614) = 2706$ ‹kJ mol^{-1}› ✓ *Bond formation*: $(6)(C–H) + (1)(C–C)$ ***OR*** $(6)(414) + (1)(346) = 2830$ ‹kJ mol^{-1}› ✓ ‹$+2706 - 2830$› $= -124$ ‹kJ mol^{-1}› ✓	*Award **[2 max]** for +124 ‹kJ mol^{-1}›.* *Award **[3]** for correct final answer.*	**3**

Question			Answers	Notes	Total
6.	**a**	**i**	<table><tr><th></th><th>Lewis (electron dot) structure</th></tr><tr><td>Ozone</td><td> ✓</td></tr><tr><td>Sulfur hexafluoride</td><td>  ✓</td></tr></table>	*Lines, x's or dots may be used to represent electron pairs.* *Charges may be included in Lewis structures of ozone but are not required.*	**2**
	a	**ii**	<table><tr><th></th><th>Electron domain geometry</th><th>Molecular geometry</th></tr><tr><td>Ozone</td><td>trigonal/triangular planar</td><td>v-shaped/bent/angular ✓</td></tr><tr><td>Sulfur hexafluoride</td><td>octahedral/square bipyramidal</td><td>octahedral/square bipyramidal ✓</td></tr></table>	*Award **[1 max]** for either both electron domain geometries correct **OR** for either both molecular geometries correct.*	**2**
	a	**iii**	sulfur hexafluoride/SF_6 ✓		**1**
	a	**iv**	*Ozone: Accept any angle greater than 115° but less than 120°* **and** *Sulfur hexafluoride:* 90° (and 180°) ✓	*Experimental value of bond angle in O_3 is 117°.*	**1**

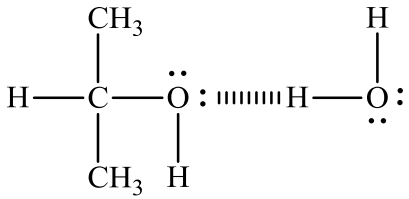
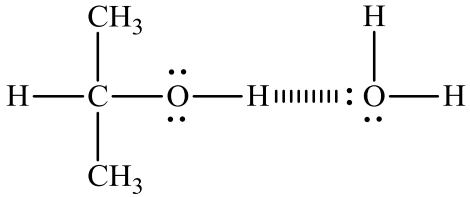
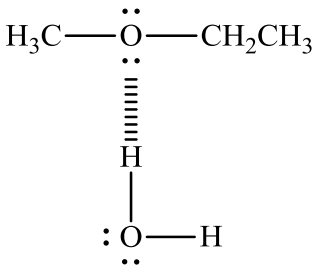
Question			Answers	Notes	Total
6.	**a**	**v**	 ✓	*Double-headed arrow not necessary for mark.* *Lines, x's or dots may be used to represent electron pairs.*	**1**
	b	**i**	<table><tr><th></th><th>Lewis (electron dot) structure</th><th>FC of O on LHS</th><th>FC of central N</th><th>FC of N on RHS</th></tr><tr><td>A</td><td>:Ö═N═N̈:</td><td>0</td><td>+1</td><td>–1</td></tr><tr><td>B</td><td>:Ö—N≡N:</td><td>–1</td><td>+1</td><td>0</td></tr><tr><td>C</td><td>:O≡N—N̈:</td><td>+1</td><td>+1</td><td>–2</td></tr></table>	*Award **[2]** for all nine FCs correct, **[1]** for six to eight FCs correct.*	**2**
	b	**ii**	smallest FC difference for A or B, so either is preferred ✓ however B is preferred as oxygen is more electronegative than nitrogen, even though FC per se ignores electronegativity ✓	*Reason required for M1.* *OWTTE*	**2**
	c	**i**	$CH_4(g) + 5O_2(g) \rightarrow CO_2(g) + 2H_2O(g) + 2O_3(g)$ ✓		**1**
	c	**ii**	$\Delta H^{\ominus} = \langle [(-393.5) + (2)(-241.8) + (2)(+142.3)] - [(-74.0)] = \rangle -660.8 \langle \text{kJ mol}^{-1} \rangle$ ✓		**1**
	c	**iii**	standard enthalpy change of formation/ $\Delta H_f^{\ominus}$ of an element ‹in most stable form› is always zero ✓		**1**
	c	**iv**	$\Delta S^{\ominus} = \langle [(+213.8) + (2)(+188.8) + (2)(+237.6)] - [(+186) + (5)(+205.0)] = \rangle$ $-144.4 \langle \text{J K}^{-1} \text{ mol}^{-1} \rangle$ ✓		**1**
	c	**v**	$\Delta G^{\ominus} = \left\langle \Delta H^{\ominus} - T\Delta S^{\ominus} = (-660.8) - (298)\left(\frac{-144.4}{1000}\right) = \right\rangle -617.8 \langle \text{kJ mol}^{-1} \rangle$ ✓		**1**
	c	**vi**	spontaneous since negative $\Delta G^{\ominus}$ ✓		**1**

Question			Answers	Notes	Total
6.	**d**	**i**	O_2 has a double bond ✓ O_3 has intermediate bonds between double and single bonds ***OR*** O_3 has a bond order of 1½ ✓ bond in O_2 is stronger therefore **I** needs more energy ✓	*Do not award mark for **I** on its own with no justification.*	**3**
	d	**ii**	C–Cl ‹bond› breaks since weakest bond ✓ $CCl_2F_2 \xrightarrow{\langle h\nu \rangle} \bullet CClF_2 + Cl\bullet$ ✓ $Cl\bullet + O_3 \rightarrow ClO\bullet + O_2$ ✓ $ClO\bullet + O\bullet \rightarrow O_2 + Cl\bullet$ ✓ $ClO\bullet + O_3 \rightarrow Cl\bullet + 2O_2$ ✓	*Allow representation of radicals without • as long as consistent throughout.*	**5**

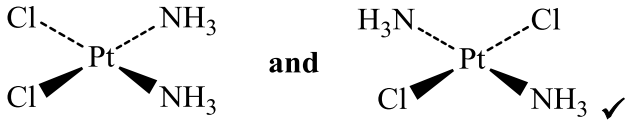
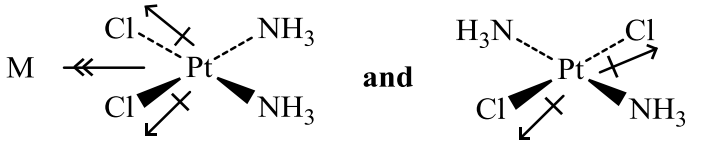
Question			Answers	Notes	Total
7.	**a**		*I:* carboxamide ✓ *II:* phenyl ✓ *III:* carboxyl / carboxy ✓ *IV:* hydroxyl ✓	*Award **[2]** for all four correct, **[1]** for two or three correct.* *Do not allow benzene.* *Do not allow carboxylic/alkanoic acid.* *Do not allow alcohol or hydroxide.*	**2 max**
	b	**i**	$n_C : \left\langle \frac{73.99}{12.01} \right\rangle = 6.161\,(\text{mol})$ **and** $n_H : \left\langle \frac{6.55}{1.01} \right\rangle = 6.49\,(\text{mol})$ **and** $n_N : \left\langle \frac{9.09}{14.01} \right\rangle = 0.649\,(\text{mol})$ **and** $n_O : \left\langle \frac{10.37}{16.00} \right\rangle = 0.6481\,(\text{mol})$ ✓ $n_C : n_H : n_N : n_O = 9.5 : 10 : 1 : 1$ ✓ *Empirical formula*: $C_{19}H_{20}N_2O_2$ ✓	*Award **[2 max]** for correct final answer without working.*	**3**
	b	**ii**	$C_{19}H_{20}N_2O_2$ ✓		**1**
	b	**iii**	$\langle (0.5)(40-20-2) = \rangle 9$ ✓		**1**
	b	**iv**	*A:* C–H **and** *B:* C=O ✓		**1**
	b	**v**	O–H **and** N–H ✓ frequencies/stretches due to O–H and N–H occur above 3200 ‹cm^{-1}› which are not present in IR of *bute* ✓		**2**
	c	**i**	1:1:6 ✓		**1**

Question			Answers	Notes	Total
7.	**c**	**ii**	 ✓		**1**
	c	**iii**	$CH_3OCH_2CH_3$ ✓		**1**
	c	**iv**	*Similarity:* both have fragment corresponding to $(M_r - 15)^+ / m/z = 45$ ✓ *Difference:* **X** has fragment corresponding to $(M_r - 17)^+ / m/z = 43$ ***OR*** **X** has fragment corresponding to $(M_r - 43)^+ / m/z = 17$ ***OR*** **Y** has fragment corresponding to $(M_r - 31)^+ / m/z = 29$ ***OR*** **Y** has fragment corresponding to $(M_r - 29)^+ / m/z = 31$ ✓	*Allow "both have same molecular ion peak/M^+ / both have m/z = 60". However in practice the molecular ion peak is of low abundance and difficult to observe for propan-2-ol.*	**2**

(Question 7 continued)

Question			Answers	Notes	Total
	c	**v**	both **X** and **Y** will exhibit hydrogen bonding with water molecules ✓ diagrams showing hydrogen bonding ✓ X:  **OR**  Y: 		**2**

(Question 7 continued)

Question			Answers	Notes	Total
7.	**d**	**i**	 ✓ Cisplatin Transplatin	*Names of complexes are not required.* *Complexes may be drawn without tapered bonds.*	**1**
	d	**ii**	 M = 0 ✓ Cisplatin Transplatin *Cis:* polar **and** *trans:* non-polar ✓		**2**
	d	**iii**	X-ray crystallography ✓	*Accept NMR spectroscopy.*	**1**
	d	**iv**	*Similarity:* both involve shared pair of electrons / both are covalent ✓ *Difference:* *Pt–N:* pair of electrons comes from nitrogen / coordinate bond **and** *N–H:* one electron comes from each bonded atom ✓		**2**
	d	**v**	London / dispersion / instantaneous induced dipole-induced dipole ✓ dipole-dipole ✓ hydrogen bonding ✓	*Award **[2]** for all three correct, **[1]** for any two correct.*	**2 max**